IIM Sirmaur Cockpit: Product Requirements Document

IIM Sirmaur | Smart Campus 2035 | BBA Operations Management, Project 4 | Version 2.0 | 2 November 2026 | Replaces "Smart Campus LMS PRD v1.0"

1. Why this version exists

Version 1.0 left two things unclear: whether the LMS and the admin dashboard are one product or two, and what "LMS" means. This version settles both. Everything below is one product, called the **Cockpit**.

#	Decision	What it means
D1	One product	One web address, one sign-in, one home screen for students, faculty, mess managers, wardens, technicians, desk staff and admins.
D2	The LMS is independent	**Learn** is a real learning management system (courses, content, assignments, grades). It owns its data and deployment and can run on its own. It joins the Cockpit through five contracts (section 7.3). This reverses v1.0 assumption 1.
D3	Campus services are a second module family	Ticketing (Control Room), mess, parcels, alerts, energy and water, student services (v1.0 M1 to M8) form the **Campus** workspace.
D4	Admin is a mode, not a second app	The **Operations console** lives inside the same shell. Eligible staff switch to it from the main dashboard with a step-up login (section 6).
D5	One identity, many roles	A person has one account and may hold several roles (for example a warden who is also faculty). Permissions are the union of roles, limited by scope.

#	Decision	What it means
D6	Humans in the loop	AI and rules recommend. Staff decide and approve (Industry 5.0).
D7	Honest data	Real, simulated and projected figures are labelled differently everywhere.
D8	Targets are proposals	2035 figures are proposed targets, not predictions. The tech stack is a recommendation.

2. Product overview

Vision. One cockpit for the whole campus: learn, get problems fixed, eat well, stay informed and, for those who run the campus, see and steer everything from one place.

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                IIM SIRMAUR COCKPIT (one shell, one sign-in)
+---------------+------------------------+-----------------------------+
| LEARN         | CAMPUS                 | OPERATIONS (admin mode)     |
| Independent   | Tickets, mess, parcels | Dashboards, KPIs, routing,  |
| LMS           | alerts, services       | SLA, users, reports, audit  |
+-------+-------+-----------+------------+--------------+--------------+
        +--- Identity | Directory | Event bus | Notifications | Audit ---+
```

Workspace	Who sees it	Purpose
Learn	Students, faculty, academic office	Courses, content, assignments, grades, attendance, timetable
Campus	Everyone	Report and track issues, mess, parcels, alerts, services and FAQ
Operations (admin mode)	Eligible staff only, after step-up login	Monitor KPIs, configure rules, manage users, publish alerts, export reports

What "independent" means for Learn.

1. It has its own database schema, API and deployment unit. Campus modules can be down and Learn still works, and the reverse.

2. It can run standalone at its own address using the shared identity service, with no Campus module installed.
3. It touches the rest of the Cockpit only through the five contracts in 7.3. It never reads Campus tables directly.
4. Removing it must not break the shell. The workspace switcher simply hides Learn.

3. Evidence, goals and metrics

Evidence base (from v1.0). Primary survey of 42 students and staff, 18 to 30 October 2026 (88% BMS, so the sample is narrow). Plus interviews, real electricity and water bills, and secondary benchmarks. Percentages are out of 42 and multi-select.

Area	Key survey baseline
Mess	Quality varies 88%, queues 67%, food runs out 50%, food wasted 48%, 57% buy food elsewhere
Hostel and repairs	Slow or incomplete 81%, problems repeat 69%, no timeline 62%, no closure confirmation 31%
Student services	Slow response 64%, repeated follow-ups 52%, 50% contact 2 or more offices per query
Resilience	About 64% hit by disruptions 2 or more times in six months, 62% say updates are unclear
Reporting habit	55% call or message, only 14% use an online portal, 12% do not report

The survey had no questions about teaching and learning tools. Learn targets are therefore proposals, and the first pulse survey adds an academic question set (section 12).

Goals.

ID	Goal
G1	Single front door: every campus problem enters through one system.

ID	Goal
G2	Visible timelines: every request shows status, owner and expected resolution time.
G3	Confirmed closure: nothing closes silently.
G4	Less mess waste and shortage through pre-booking and demand forecasting.
G5	Clear communication in disruptions, with acknowledgement tracking.
G6	Evidence for decisions: admins see live KPIs against the survey baseline.
G7	One login, one home: academic and campus life in the same place.
G8	Secure, fast admin access: eligible staff reach the Operations console from the main dashboard in under 20 seconds, with every use audited.

Success metrics (baseline and proposed 2035 target).

Metric	Baseline	Target
Repairs reported slow or incomplete	81%	under 20%
Problems that repeat (same asset and category in 30 days)	69%	under 20%
Open tickets with no expected resolution time	62%	near 0%
Tickets closed without confirmation or auto-close notice	31%	0%
Unclear disruption updates	62%	under 15%
Students eating elsewhere because of the mess	57%	under 20%
Students visiting 2 or more offices per query	50%	under 10%
Food runs out / food wasted	50% / 48%	under 10% / under 15%

Metric	Baseline	Target
Weekly active use of Cockpit by students (new, no baseline)	not measured	over 80%
Courses actively using Learn (new, no baseline)	not measured	over 90%
Admin switch completed in under 20 seconds (new)	not measured	95% of attempts

Non-goals. Replacing ERP, finance or admissions systems. Payment processing. Biometrics or facial recognition. Controlling real hardware in the prototype (sensor data is simulated). Replacing staff.

4. Users, roles and workspaces

ID	Role	Learn	Campus	Can switch to admin mode?
R1	Student (resident or day scholar)	Learner	Requester, rate meals, parcels, alerts	No
R2	Faculty and staff	Instructor	Requester, alerts	No (instructor tools live in Learn)
R3	Mess manager	Optional	Staff tools: menu, plan, waste log	Yes, Mess console
R4	Hostel warden	Optional	Staff tools: block queue, assign	Yes, Hostel console
R5	Maintenance technician or vendor	No	Staff tools: job list only	No
R6	Service desk officer	No	Staff tools: desk queue, FAQ	Yes, Services console

ID	Role	Learn	Campus	Can switch to admin mode?
R7	Parcel desk or security	No	Staff tools: parcel log, verify	Yes, Parcels console
R8	Operations admin (Director of Operations)	Read-only view	Everything in scope	Yes, full console
R9	System administrator	Platform settings	Platform settings	Yes, full console plus system settings
R10 (new)	Academic office	Manage courses, terms, timetable, enrolment	Requester	Yes, Academic console

Two kinds of staff screens.

- **Staff tools** are daily work screens (work queue, job list, parcel log, waste log). They open with normal sign-in because they are used many times a day and affect only the person's own scope.
- **The Operations console** covers cross-cutting views, configuration, publishing alerts, exports and user management. It needs admin mode.

5. The Cockpit shell and dashboard

Global elements (always visible). Workspace switcher (Learn, Campus, and Operations only for eligible users). Global search across courses, tickets, FAQ and menu. Live alert banner. Notification centre with unread count. Profile menu with the **Switch to Admin mode** entry for eligible users only. Language setting (English first, Hindi later).

Home cockpit by role. The home screen is a set of cards. Each role sees the cards it needs.

Role	Home cards
Student	Large Report an issue button, active tickets, today's classes and next deadlines (from Learn), today's mess menu with Skip buttons, parcels ready, live alerts
Faculty	Today's classes, submissions to grade, announcements to post, my tickets, live alerts
Mess manager	Tomorrow's pre-booked demand and forecast, today's ratings, waste and shortage, open quality tickets
Warden	Block queue with countdowns, overdue tickets, repeat-issue hotspots, live alerts
Technician	Today's job list by scheduled window
Director and admin	Switch to Admin mode card with the number of items needing attention, plus the same personal cards

Admin mode look and behaviour.

- A coloured top band reads "ADMIN MODE" with the role, scope, a session countdown and an **Exit admin mode** button.
- The left navigation changes to the Operations console views the person is allowed to see.
- Colour is never the only signal (accessibility). The band also carries text and an icon.
- Returning to the personal dashboard never asks for credentials.

6. Admin switch: authentication and session design

This is the key integration requested for the Cockpit. A signed-in user reaches the admin side from the main dashboard, without logging out, by proving who they are a second time.

6.1 Two levels of sign-in.

Level	How	Gives access to
Level 1: standard sign-in	Institute email and a one-time code (10-minute expiry, 5 attempts) or institute single sign-on	Personal workspace, Learn, Campus, staff tools
Level 2: admin step-up	Admin passphrase plus a second factor (authenticator code, or email code in the prototype)	Operations console for the roles the user holds

6.2 Who can switch. Only users with an active admin-eligible role assignment (R3, R4, R6, R7, R8, R9, R10). Everyone else never sees the entry. Eligibility is granted by the system administrator and reviewed every term. Admin passphrases are created at onboarding through a one-time invite and are separate from the sign-in method. Production target: institute single sign-on with multi-factor authentication replaces the separate passphrase. This is an open decision (section 13).

6.3 Flow S1: switching to admin mode.

Step	Actor	What happens
1	User	On any screen, opens the profile menu or home card and taps **Switch to Admin mode**.
2	System	Checks eligibility and that the account is not locked. Shows a panel stating which role and scope the user will operate under.
3	User	Enters the admin passphrase.
4	User	Enters the second-factor code.
5	System	Verifies both. Issues a short-lived admin token (mode=admin, role and scope claims, expiry). Writes the audit event admin.mode_entered with time and device.
6	System	Transforms the shell: admin band, countdown, Operations navigation. Opens the default console view for the role (Today for R8, Mess for R3, Hostel for R4, Services for R6, Parcels for R7, Academic for R10).

Step	Actor	What happens
7	User	Works. Sensitive actions ask for the second factor again (see 6.4).
8	User or system	**Exit admin mode** or timeout revokes the token, returns the user to the same personal screen where possible, and writes admin.mode_exited.

6.4 Session rules (proposed values, stored as settings).

- Idle timeout 30 minutes, maximum session 4 hours. A warning appears 2 minutes before expiry with a one-tap extend.
- Re-prompt for the second factor on sensitive actions: exporting personal data, changing roles or permissions, editing SLA or routing rules, publishing an emergency alert, deleting records.
- Five failed attempts lock admin switching for 15 minutes and notify the system administrator. Failed attempts never lock Level 1 access.
- One active admin session per person. Opening a second one ends the first.
- Duty wardens can authenticate at shift start so an emergency alert is never delayed by login.
- Admin mode never opens by URL alone. A deep link to an Operations view shows the step-up panel first.

6.5 Authorization design.

1. Every API call carries role, scope and mode claims. Operations endpoints reject tokens without mode=admin.
2. Row-level rules in the database enforce role and scope. Hiding a button is never the only protection.
3. A user holding several roles gets the union of consoles, each still limited by its own scope.
4. If eligibility is revoked mid-session, the next API call fails and the shell drops to the personal workspace within 60 seconds.

6.6 Audit. The log records every switch, failure, timeout and sensitive action with who, what, when and device. Entries cannot be edited. Only R8 and R9 can read the log.

6.7 Edge cases. Lost authenticator: the system administrator resets it with a second approver. Shared or public computers: admin mode shows a warning and uses the shorter timeout of 10 minutes. Mobile: the step-up panel is a single screen with large inputs. User with no admin eligibility who guesses the URL: shown a plain "not available for your account" page and an audit event.

6.8 Acceptance criteria.

ID	Requirement	Priority	Acceptance criterion
S-01	Switch entry only for eligible users	Must	A student account shows no switch entry and gets a refusal from the admin API
S-02	Step-up with passphrase and second factor	Must	Wrong passphrase or code never opens admin mode. Correct values open it in under 20 seconds
S-03	Visible admin state	Must	Band, badge and countdown appear on every admin screen
S-04	Timeouts	Must	Idle session ends at 30 minutes and API calls then return unauthorized
S-05	Exit without logout	Must	Exit returns to the personal dashboard with no credentials requested
S-06	Lockout	Should	Five failures lock switching for 15 minutes and notify the sysadmin
S-07	Sensitive-action re-prompt	Should	Role change, export and SLA edit each require a fresh code
S-08	Audit of the whole life cycle	Must	Entered, exited, timed out, failed and sensitive events all appear in the log
S-09	Scope enforcement in admin mode	Must	A Block B warden sees no Block A data even in admin mode
S-10	Revocation	Should	Removing eligibility ends the session within 60 seconds

7. Learn: the independent LMS

7.1 Modules.

ID	Module	Priority	Users	Purpose
L0	Learn core	Must	All	Terms, programmes, courses, sections, enrolment, roles
L1	Course space	Must	R1, R2	Weekly structure, files, links, video, readings
L2	Assignments	Must	R1, R2	Create, submit, grade, feedback, late rules
L3	Gradebook	Should	R1, R2, R10	Marks, weights, grade export, student view of own marks
L4	Quizzes	Should	R1, R2	Question bank, timed quizzes, auto-grading for objective items
L5	Attendance	Should	R1, R2, R10	Session register by instructor, summary for students
L6	Announcements and discussion	Must	R1, R2	Course notices, threaded Q and A
L7	Timetable and calendar	Must	R1, R2, R10	Weekly timetable, exam dates, deadlines, one calendar feed
L8	Learning insights	Could	R2, R10	Participation and late-submission patterns shown to instructors, who decide any action

7.2 Functional requirements.

ID	Requirement	Priority	Acceptance criterion
L-01	Enrolment lists loaded by the academic office per course	Must	A student sees only the courses they are enrolled in
L-02	Instructor creates a course space with weekly sections	Must	A new course page can be made and published in under 5 minutes

ID	Requirement	Priority	Acceptance criterion
L-03	Upload files up to 50 MB and add links	Must	A 20 MB PDF uploads and opens on a phone
L-04	Assignment with due date, instructions and attachments	Must	Students are notified at creation and 24 hours before the due date
L-05	Student submission with receipt	Must	A submission shows a timestamp, and late work is flagged
L-06	Instructor grading with written feedback	Must	The student sees marks and feedback, and cannot see other students'
L-07	Course announcements with notifications	Must	A post reaches all enrolled students within 10 seconds in-app
L-08	Discussion threads per course	Should	Instructors can pin, resolve and moderate
L-09	Quiz with timer and auto-grading	Should	Objective questions are scored on submission
L-10	Attendance register per session	Should	A student sees attendance percentage per course
L-11	Timetable view and calendar feed	Must	Rooms and times appear on the student home screen
L-12	Gradebook export	Could	Instructors download a CSV, never marks of other courses
L-13	Standalone mode	Must	With Campus modules switched off, Learn signs in and works fully

7.3 Integration contracts between Learn and the Cockpit.

ID	Contract	Direction	Content
C1	Identity	Cockpit to Learn	Signed token with user id, roles and scope. Learn keeps no passwords
C2	Directory	Cockpit to Learn	Programmes, locations (rooms), people. Learn caches it

ID	Contract	Direction	Content
C3	Events	Learn to Cockpit	assignment.created, assignment.due_soon, grade.published, announcement.posted, session.scheduled
C4	Calendar feed	Learn to Cockpit	Read-only list of classes and deadlines per user for the home screen
C5	Notifications	Learn to Cockpit	Learn asks the shared notification service. It does not send its own messages

8. Campus workspace

The Campus modules from v1.0 are carried forward, including all flows F0 to F9, the ticket state machine and the acceptance criteria in v1.0 sections 8 and 9, unless the changes below apply.

ID	Module	Priority	Release	Change in v2.0
M0	Platform core (identity, roles, notifications, audit)	Must	R1	Adds admin step-up (section 6)
M1	Control Room ticketing	Must	R1	Adds "Report an issue" from a course page with room prefilled from the timetable
M2	Mess	Must	R1	Adds class-clash check for meal timings using the timetable
M3	Hostel and infrastructure	Should	R2	No change
M4	Student services and knowledge base	Should	R2	Adds academic FAQ owned by the academic office
M5	Resilience and alerts	Should	R2	Banner appears in Learn as well
M6	Parcel and delivery	Could	R3	No change
M7	Energy and water	Could	R3	No change
M8	Campus pulse	Could	R3	Adds academic question set

Core rules kept from v1.0.

- Every request is a ticket with category, location, owner, priority and timeline. States: Submitted, Acknowledged, In progress, On hold, Resolved, Closed, Reopened, Auto-closed, Cancelled, Merged.
- Resolution needs requester confirmation. After 72 hours with no reply (reminders at 24 and 48), it auto-closes and is flagged "unconfirmed".
- Escalation levels: L0 owner group, L1 supervisor, L2 operations admin, L3 Director of Operations. Escalation is a counter, not a state.
- Three repeats on the same asset or location in 90 days create a Problem record needing a root-cause note.

Priority	Example	Acknowledge	Resolve
P1 Critical	Hostel power failure, water outage, safety hazard, food-safety incident	15 minutes	4 hours
P2 High	No hot water, broken lock, blocked drain	1 hour	24 hours
P3 Normal	Fan or light not working, furniture	4 hours	72 hours
P4 Low	Cosmetic repairs, suggestions	24 hours	7 days

These values are proposals and must be checked with wardens and maintenance staff.

9. Operations console (admin mode)

Principles. Decisions first (the first screen answers "what needs attention now?"). Every number drills down to its records. Real, simulated and projected data are labelled. Filters for date, block, category, owner group and priority work across views. Every table exports to CSV and every view prints to PDF. KPIs show baseline, current value and target.

View	Main content	Mess mgr	Warden	Desk	Parcel	Academic	Ops admin	Sysadmin
Today	P1 and P2 tickets, overdue, L2 and L3, live alerts, yesterday's mess shortage, new anomalies	Mess only	Block	Office	Parcels	Academic	All	All
Control Room board	Tickets by state with countdowns, assign, escalate	Scope	Block	Office	Scope	Scope	All	All
Mess	Forecast vs actual, waste, shortage, ratings, queue time	Yes	No	No	No	No	Yes	View
Hostel and infrastructure	Backlog, time to fix, repeat hotspots, Problem records	No	Block	No	No	No	Yes	View

View	Main content	Mess mgr	Warden	Desk	Parcel	Academic	Ops admin	Sysadmin
Student services	Volume, response time, transfers, FAQ deflection, stale articles	No	No	Yes	No	Yes	Yes	View
Energy and water	Consumption against real bills, anomalies, savings	No	No	No	No	No	Yes	View
Parcels	Median days to collect, missing cases, unclaimed	No	No	No	Yes	No	Yes	View
Alerts and incidents	Alerts, reach, durations, review notes	No	Block	No	No	No	Yes	Yes
Academic operations (new)	Courses without content, assignments overdue for grading, enrolment gaps, timetable clashes	No	No	No	No	Yes	View	View
KPI scorecard	All KPIs against baseline and target	No	No	No	No	No	Yes	View
Reports	Weekly summary and exports	Mess	Block	Office	Parcels	Academic	All	All
Settings	Users and roles, categories, SLA, routing, locations, assets, FAQ, audit log	No	No	No	No	Terms, courses	View	Yes

New KPIs.

KPI	Definition	Refresh
Learn weekly active rate	Students who opened Learn in the week divided by enrolled students	Daily
Assignment feedback time	Median days from due date to grade published	Daily

KPI	Definition	Refresh
Courses with content	Courses with at least one item published in the last 14 days divided by active courses	Daily
Admin switch time	Median seconds from tapping Switch to the console loading	Daily
Admin session audit exceptions	Count of failed attempts, lockouts and timeouts	Hourly

All v1.0 KPIs (first response time, resolved within target, closure confirmation, repeat rate, reopen rate, escalation rate, skip rate, forecast error, waste share, alert reach and others) are kept with the same definitions.

10. Cross-module flows

ID	Flow	Steps	Value
X1	Admin switch	See section 6.3	Secure access to the console from the main dashboard
X2	Assignment to home screen	Instructor posts an assignment. Learn emits assignment.created. The student home deadline card and notification update	One place for deadlines
X3	Classroom issue from a course	A student or instructor taps Report an issue in a course. The room is prefilled from the timetable. A ticket is created and routed	Fewer calls and messages
X4	Meal and class clash	The mess module compares meal slots with the timetable and flags clashes to the manager and academic office (19% report clashes)	Less time lost

ID	Flow	Steps	Value
X5	Disruption alert	A published alert shows as a banner in Learn and Campus. Instructors get a prompt to consider adjusting deadlines. They decide	Clear updates (62% say unclear)
X6	Dashboard drill-down	An admin sees a spike in overdue tickets, opens it, reassigns and adds a note without leaving admin mode	Evidence to action

11. Architecture, data and security

Layers.

Layer	Components
Experience	One Next.js shell with route groups: Personal, Learn, Campus, Staff tools, Operations. Progressive web app for phones
Shared services	Identity and roles, admin step-up service, directory, event bus, notification service, audit log, search
Learn service	Own schema, API, file storage
Campus services	Ticket engine, SLA and escalation, routing, mess, parcels, alerts, utilities, knowledge base, analytics
Intelligence	Demand forecasting (exponential smoothing and the newsvendor rule), anomaly detection, duplicate and repeat-issue detection
Data	PostgreSQL with separate schemas (core, lms, campus), time-series readings, object storage for photos and files

Prototype stack (recommendation). Next.js with Tailwind, Supabase (PostgreSQL, authentication, row-level security, storage), Recharts, a small Python service for forecasting, Vercel hosting. The same architecture holds if the team prefers another stack.

New data entities.

Entity	Key fields
AdminEligibility	user, console, scope, granted by, start and end, reviewed on
AdminSession	user, started, last activity, expiry, device, exit reason (manual, timeout, revoked)
StepUpAttempt	user, time, result, reason for failure
Course, Section, Enrolment	term, programme, instructor, student, status
ContentItem, Assignment, Submission, Grade	course, type, due date, file, marks, feedback
Session, AttendanceMark	course, date, room, student, present or absent
Announcement, Thread, Post	course, author, body, pinned

All v1.0 entities (User, RoleAssignment, Location, Asset, Category, Ticket, TicketEvent, MealSlot, MealBooking, MealRating, WasteLog, ForecastRun, Parcel, Alert, Meter, Anomaly, KBArticle and the rest) are kept.

Security and privacy.

- Only institute accounts can sign in. Row-level security enforces role, scope and mode in the database.
- Student marks, attendance and tickets are visible only to the student, their instructors or owners, and authorised staff.
- Meal ratings can be anonymous. Staff see aggregates and text only.
- Photos are stored privately with location metadata stripped.
- Collect only name, institute email, role, room or office, and an optional phone number. No biometrics.
- Retention (proposed): tickets three years, photos one year, grades and course records per institute policy, audit log five years.

- Designed to be consistent with India's Digital Personal Data Protection Act, 2023. The final policy needs confirmation from the institute's data owner.

Non-functional requirements.

Area	Requirement
Performance	App loads in under 3 seconds on 4G. Ticket submits in under 2 seconds. Dashboards load in under 3 seconds from precomputed figures
Scale (assumed)	2,000 users, 500 tickets per day, 100,000 meter readings per day. Confirm real population before the demo
Availability	99% in semester, maintenance outside meal times and late evenings
Offline	Home screen, menu, timetable and tickets cached. Creating a ticket works offline and syncs later
Accessibility	WCAG 2.1 AA, tap targets at least 44 pixels, readable at 200% zoom
Usability	4 of 5 test students complete a ticket unaided in under 90 seconds

12. Release plan, demo and risks

Releases.

Release	Scope
C1 (demo-ready)	Shell, single sign-in, admin step-up (section 6), M0, M1, M2, Operations views Today, Control Room, Mess and KPI scorecard, Learn minimum (L0, L1, L2, L6, L7)
C2	M3, M4, M5, Learn L3 to L5, Academic operations view
C3	M6, M7, M8, Learn L8, Production single sign-on

Five-week build for C1.

Week	Focus	Output
1	Foundations	Repository, schemas (core, lms, campus), sign-in, roles, seed data, shell with workspace switcher
2	Admin switch and ticketing	Section 6 flow S1 working, ticket creation, routing, SLA timers
3	Staff and Learn	Work queue, technician list, closure and escalation, Learn course page, assignment and announcement
4	Mess and dashboard	Menu, skip, rating, forecast, waste log, Operations views
5	Test and demo	Usability test, simulated data labelled, backup recording, rehearsal

Demo script (about 2 minutes).

1. Sign in as a student. Show the home cockpit: next class and deadline from Learn, mess menu, live alert banner.
2. Open a course, view an assignment, then raise a plumbing ticket with a photo.
3. Switch to the warden account. Accept and assign the ticket. Switch to the technician and mark it fixed with a photo.
4. Return as the student and tap Yes, fixed.
5. As the Director, tap **Switch to Admin mode**, enter the credentials, and show the admin band. Open Today, then the KPI scorecard with survey baselines beside current values.
6. Open the Mess view for forecast against actual, waste and shortage. Tap Exit admin mode to return to the personal dashboard.

The prototype labels all sensor, footfall and seeded mess data as "Simulated". Keep a backup video of the same sequence.

Definition of done. A feature works for the roles it is written for, passes its acceptance criterion, shows loading, empty and error states, has labelled data sources and is checked by a second team member.

Risks.

Risk	Likelihood	Impact	Mitigation
Scope grows with Learn added	High	High	Build C1 only, keep C2 and C3 as designs, use feature flags
Admin switch is too heavy for daily staff work	Medium	High	Keep daily tools outside admin mode, allow shift-start authentication
Admin credentials mishandled in the demo	Medium	High	Demo accounts clearly flagged, rotated after the demo, never reused in production
Staff and faculty do not adopt	Medium	High	Few buttons, short training, instructor onboarding for Learn
Demo fails on connection	Medium	High	Backup recording, local copy
Demo data mistaken for real	Medium	Medium	Labels on every figure, legend in the script
SLA values unrealistic	Medium	Medium	Review with warden and mess manager, store as settings
Survey sample is narrow (88% BMS)	High	Medium	State the limit, widen the next round

13. Open questions for the team

1. **Learn depth.** Is the professor expecting full LMS features (quizzes, gradebook), or is the C1 minimum (courses, assignments, announcements, timetable) enough for the demo?
2. **Admin credentials.** Is a separate admin passphrase plus a second factor acceptable, or should the Cockpit use institute single sign-on with multi-factor only? v1.0 had no passwords.
3. **Who is admin-eligible?** Confirm the list in section 4, including whether duty wardens and security can use the alert tool in admin mode.
4. **Session length.** Are 30 minutes idle and 4 hours maximum right for staff?
5. **Stack.** Is Next.js with Supabase comfortable for the team, or is another stack already chosen?

6. **Mess attendance default.** Assume attending and tap Skip (recommended), or opt in per meal?
7. **Mess data.** What can the mess manager share: meals per day, attendance, quantity prepared, waste, cost per portion?
8. **Bills.** How many months of real electricity and water bills exist, and at what level (campus, block, meter)?
9. **SLA values.** Do the proposed values match what wardens and maintenance staff think is achievable?
10. **Channels and population.** Is SMS or WhatsApp available for the demo, and how many users should the demo assume?
11. **Presentation date.** This sets how the five-week plan is compressed.

Glossary

Cockpit: the single shell that holds Learn, Campus and the Operations console. **Learn:** the independent learning management system. **Admin mode:** the Operations console, opened by step-up login inside the main dashboard. **Step-up authentication:** asking for stronger proof of identity before a sensitive area opens. **Staff tools:** daily work screens that do not need admin mode. **SLA:** the promised time to acknowledge and resolve a request. **Problem record:** a record created when the same issue repeats, used to find the root cause. **Newsvendor model:** an inventory method that balances the cost of shortage against the cost of leftover stock. **Row-level security:** database rules that decide which rows each user can read or change.